UG (NEP) 5th SEMESTER
COMPUTER APPLICATIONS – DISCRETE MATHEMATICS
(CAP522J3)

Session: December 2025 No. of Questions: 09
Time Allowed: 2:00 Hours Maximum Marks: 100 Minimum Marks: 36

NOTE: Attempt all questions from Sections “A” and “B” and only two questions from Section “C”.

SECTION A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

1. For the function f(x) = 2x + 2, find the range when domain = {2, −1, 0, 1, 2}.
2. Write the difference between tautology and contradiction.
3. What is the Cartesian product of two sets? Give an example.
4. Define universal and existential quantifiers.
5. Evaluate the binomial expression $(2x + 3)^5$.
6. What is a multigraph? Give one example.
7. Define a bijective function.
8. What is a spanning tree?

SECTION B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. Show that $f(x) = 2x^2 + 3x + 1$ is $O(x^2)$.

OR

Let A = {1, 2, 3, 4} and B = {4, 5, 6}. Find A ∪ B, A′, and (A ∪ B)′ with the help of Venn diagrams.

3. Construct truth tables for: (i) P ∨ (Q ∧ R) (ii) (P → Q) ↔ (¬Q → ¬P).

OR

State and prove De Morgan’s Laws of Logic.

4. How many cards must be selected from a standard deck of 52 cards to guarantee that at least three cards of the same suit are chosen?

OR

A box contains 10 blue, 20 red, 8 green, 15 yellow and 25 white balls. How many balls must be chosen to ensure that we have 12 balls of the same colour?

5. Explain graph terminology and types of graphs.

OR

What are Euler and Hamiltonian circuits? Explain with examples.

SECTION C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. Prove by Mathematical Induction that: $2 + 6 + 10 + \ldots + (4n - 2) = 2n^2$.
7. If a graph G with n vertices has k components, then G has at most $(n - k)(n - k + 1)/2$ edges.
8. State the Pigeonhole Principle. Also state the Generalized Pigeonhole Principle. Give a suitable example.
9. Discuss in detail the Seven (7) Bridge Problem of Königsberg with a suitable example. What is the current version of this problem?